Ans-4

Data transformation techniques

Meaning

It means changing data into suitable format for analysis and modeling

Techniques

- Normalization: Scale values between 0 and 1
- Standardization: mean = 0, variance = 1
- Encoding: Convert text to numbers
- Smoothing: removes noise

Importance

- Improves model performance
- Makes data more understandable

ns-3

Handling Large Datasets Efficiently

* Techriques
  - Use efficient libraries (Pandas, Numpy)
  - Load data in small chunks insted of full data
  - Remove unecessary colums/rows
  - Avoid loops, use builtin-function
* Importance
  - Saves memory
  - Prevents system Crash

s-2

String Manipulation

It is modifying & working with text data

Example

text = "hello"

text.upper() → HELLO

text.strip() → "hello"

text.replace("hello", "hi") → hi

Importance

- Important in NLP, Chatbots, data Processing

ns-1

Data Wrangling

It is the process of cleaning, organizing and preparing raw data for analysis

It includes

- Remove incorrect or duplicate data
- Filling missing values
- Converting data into proper format

Importance

- Makes data clean & useable
- Improves accuracy of data